

DANIEL NADEEM | ELECTRICAL ENGINEER

www.danielzn.com – (647) 204-9268 – danielzn@alumni.ubc.ca – willing to relocate

SKILLS

Programming

- C/Python
- Verilog
- Git

Hardware

- Circuit Design
- Lab Equipment
- Soldering

Software Tools

- SolidWorks
- Altium
- LTSpice

Communication

- Word/Excel
- Data Processing
- WordPress

EXPERIENCE

Tesla Fremont, California

May 2022 – Aug. 2022

Test Engineering Intern

Developed HV lab test equipment and performed tests on HV powertrain products

- Designed energy-limiting circuit interruption device to protect lab equipment from overheating
 - Currently used by Tesla HV test lab to protect high-power rated resistor bank
- Created programs to control power supply and read CAN data to automate battery charging
 - Reduces routine battery test setup time for lab technicians by 2+ hours per day
- Performed HV electrical abuse tests to validate the response of powertrain components

EnerSys Burnaby, British Columbia

Sep. 2021 – Dec. 2021

Electronics Engineering Intern

Performed tests on various electronic devices using self-designed jigs and industrial lab equipment

- Designed high voltage shorting jig with an isolated switch to facilitate testing of power systems
- Successfully diagnosed faulty shunt measuring PCBA to root out cause of measurement discrepancy

Unmanned Aerial Systems UBC Vancouver, British Columbia

Sep. 2022 – May 2023

RCOMMS Sub-team Member

Designed long range directional antenna to extend range of 2.4GHz network to drone

- Determined theoretical link budget of communication system for part selection
- Running test of communication link to determine effectiveness of previously selected parts

Rocket UBC Vancouver, British Columbia

Sep. 2022 – May 2023

Avionics Sub-team Member

Designed radio communications PCBA to relay information from rocket

- Selected parts to fit requirements including size, range, and noise profile
- Designed PCB from schematic to PCB tracing with team members and feedback from design reviews
 - Used on *Beauty and the Beast* during Launch Canada 2023 competition

PROJECTS

Underground Proximity Detection Device

Sep. 2022 – May 2023

Designing a peer-to-peer RF distance measurement system to warn of close proximity

- Performed RF strength measurement tests to correlate power received to distance
- Building entire system architecture including processor, battery, and interface to make device portable

Quality Control Robot

Jan. 2021 – Apr. 2021

Designed a robotic arm with 3.5 degrees of freedom to grab marshmallows off a conveyer belt

- 3D modelled the various arm components to be compatible with off-the-shelf motors and hardware

EDUCATION

University of British Columbia

Sep. 2018 – May 2023

Bachelor of Applied Science in Electrical Engineering